

NAME: _____

PERIOD: _____

Average Payoff, Typical Ticket

AP Statistics · Unit 2 · Topic 2.8

[STAR] THE PROBLEM

Suppose a club sells 200 raffle tickets for \$2 each. One ticket wins \$100, four tickets win \$25 each, 20 tickets win \$5 each, and the other tickets win nothing. Let X be the buyer's net profit from one randomly chosen ticket: prize minus ticket cost.

Priya: *The average gross payout is \$1.50 per ticket, so buying one is a good deal.*

Mateo: *Most tickets win nothing.*

Ticket net-profit distribution

Prize	Net x	$P(X = x)$
Top prize	98	1/200
Second prize	23	4/200
Small prize	3	20/200
No prize	-2	175/200

FIRST TAKE — one sentence before you work the parts

Would you buy one ticket if your goal were to make money? Use one number or outcome from the problem.

[BOOK] DISTRIBUTIONS AND ONE OUTCOME (keep on the page)

A discrete random variable assigns a number to each random outcome. Its distribution lists every possible value with a probability between 0 and 1; all probabilities add to 1. Add the probabilities of the outcomes in an event. Net profit is prize minus ticket cost. An average can differ from the most common outcome; large prizes can raise it, and it does not promise one buyer's result.

Work (a)–(c).

DOK 1 (a) State the four possible net profits X . Which is most likely?

DOK 2 (b) Check that the listed probabilities add to 1. Find the chance that a ticket loses money.

DOK 3 (c) Can Priya's average payout and Mateo's statement both be true? Use the distribution to explain. Decide whether Priya's good-deal claim follows for a buyer trying to make money, using the ticket cost and possible net profits. Write 3–4 sentences.

Frame: Both statements can be true because the few _____ raise the average while most tickets _____.

Frame: For a buyer trying to make money, I would _____ because the ticket costs _____ and the distribution shows _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.